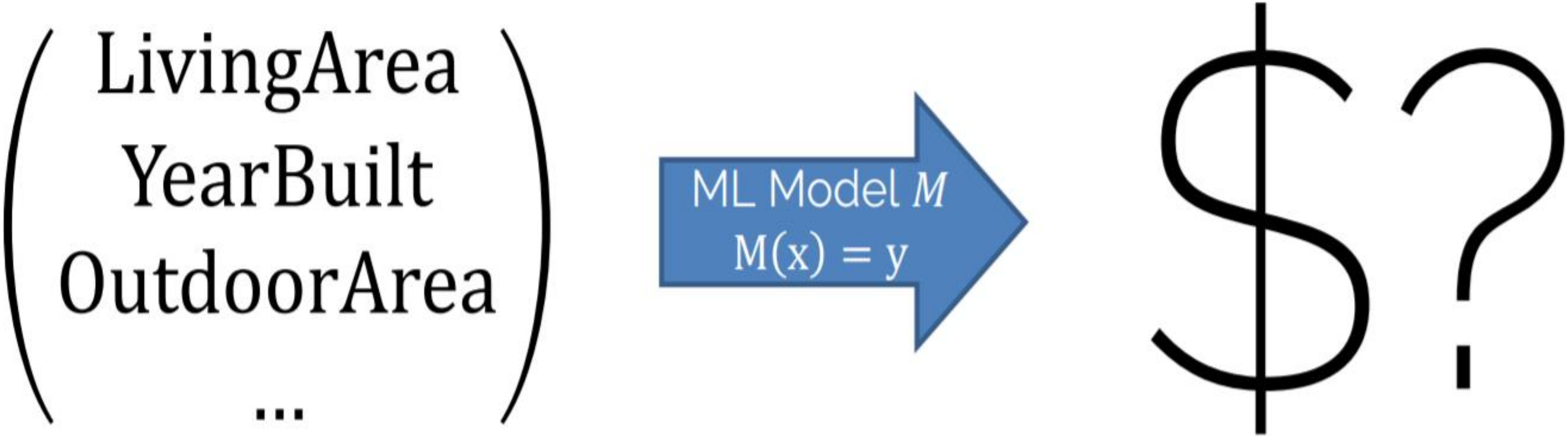
Linear Regression

Linear Regression for House Prices

$?

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Linear Regression for House Prices



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Linear Regression

Find a linear model that explains a target $\boldsymbol{y}$ given inputs $\boldsymbol{x}$

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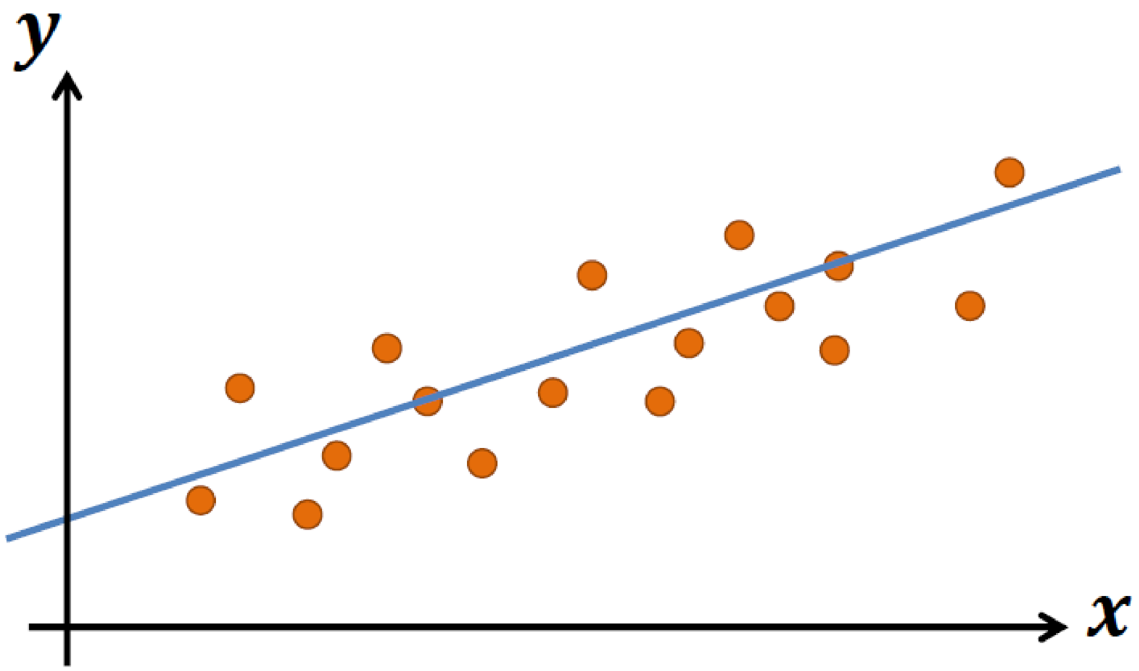

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Linear Regression

Training

$$\{\boldsymbol{x}_{1:n}, \boldsymbol{y}_{1:n}\}$$

Data points

Linear Regression

Training

$$\{\boldsymbol{x}_{1:n}, \boldsymbol{y}_{1:n}\}$$

Data points

Input (e.g., image, measurement)

Linear Regression

Training

$$\{\boldsymbol{x}_{1:n}, \boldsymbol{y}_{1:n}\}$$

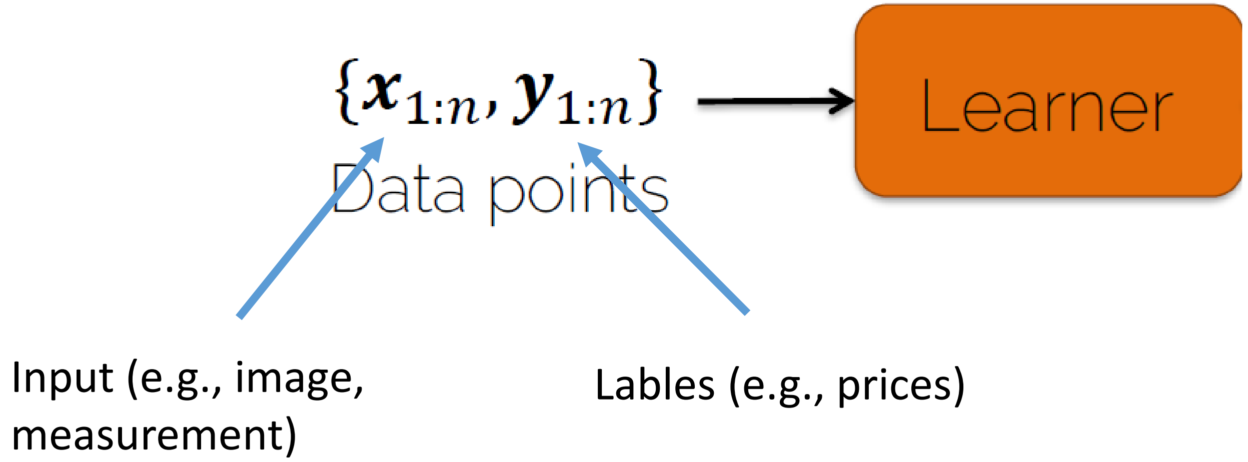
Data points

Input (e.g., image, measurement)

Lables (e.g., prices)

Linear Regression

Training



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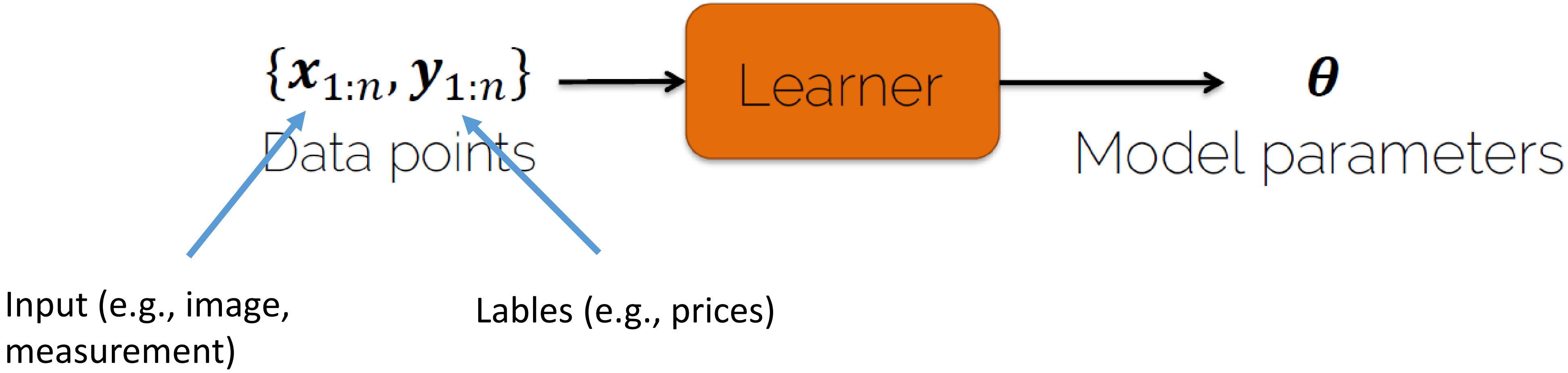
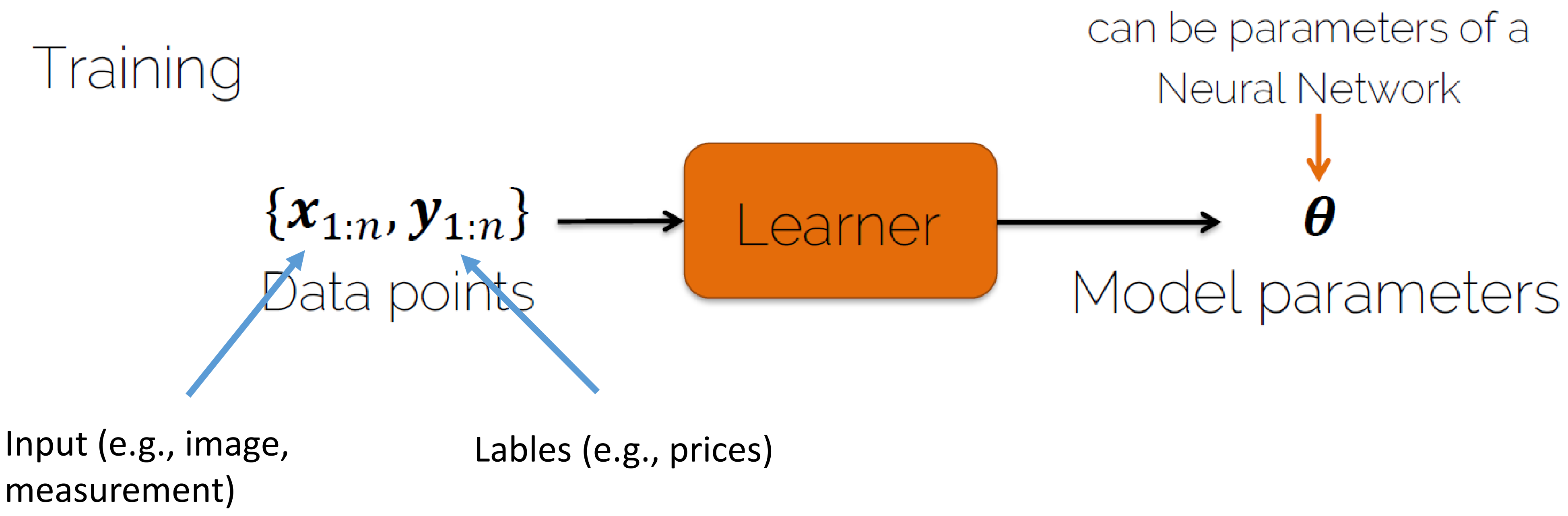
Linear Regression

Training



Linear Regression



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Linear Regression

Training

can be parameters of a
Neural Network

$$\{\boldsymbol{x}_{1:n}, \boldsymbol{y}_{1:n}\} \longrightarrow \boxed{\text{Learner}} \longrightarrow \boldsymbol{\theta}$$

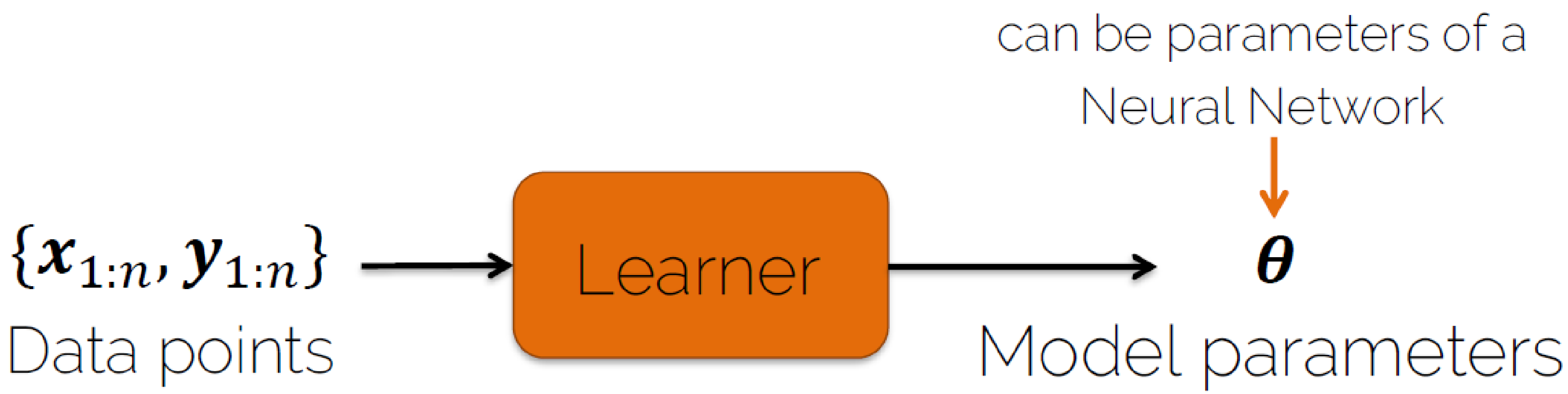
Data points

Model parameters

Testing

Linear Regression

Training

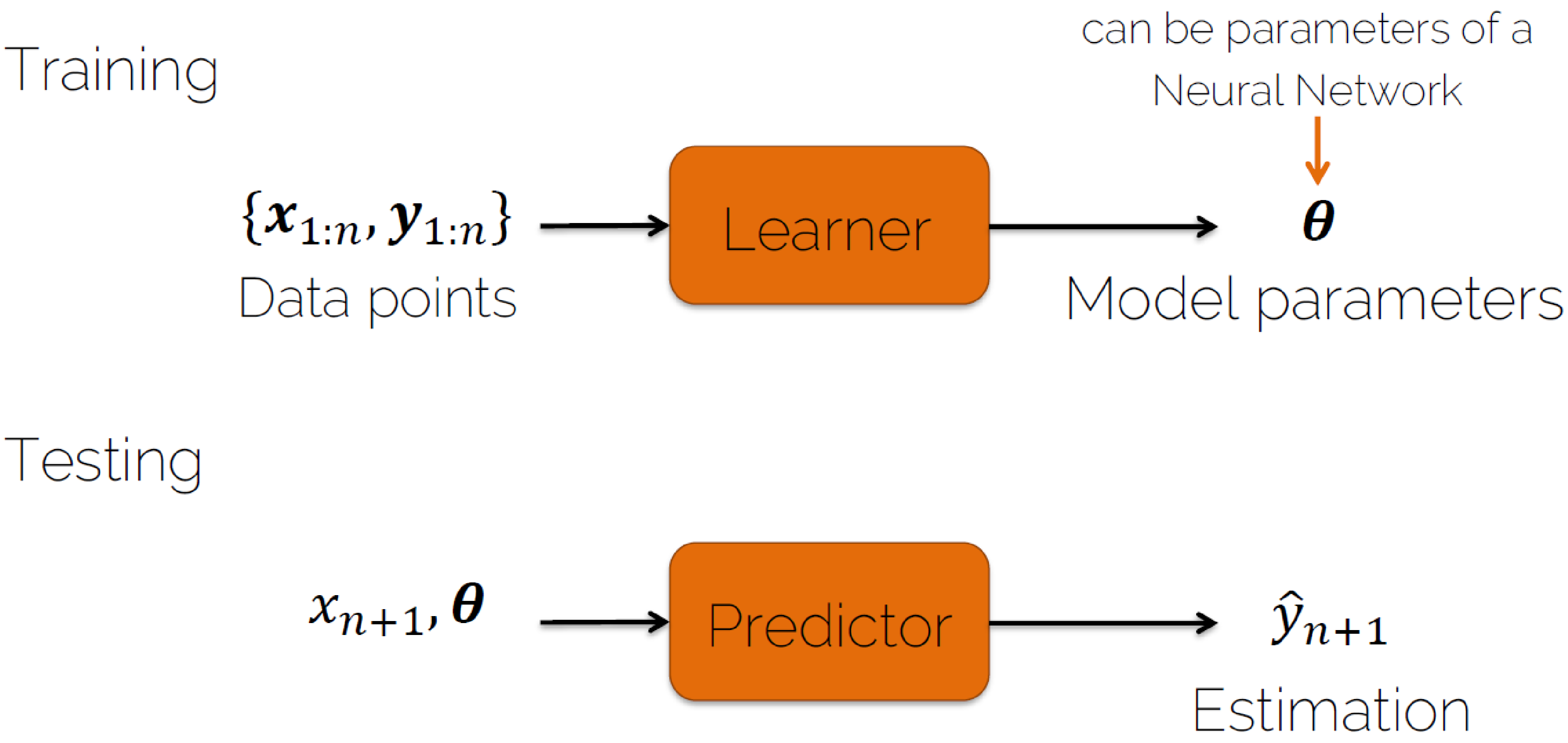


Testing

$x_{n+1}, \boldsymbol{\theta}$

Adapted from: Niessner and Leal-Taixé, Introduction to Deep Learning

Linear Regression



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Linear Prediction

A linear model is expressed in the form

$$\hat{y}_i = \sum_{j=1}^{d} x_{ij}\theta_j$$

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input data

Linear Prediction

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Linear Prediction

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input dimension

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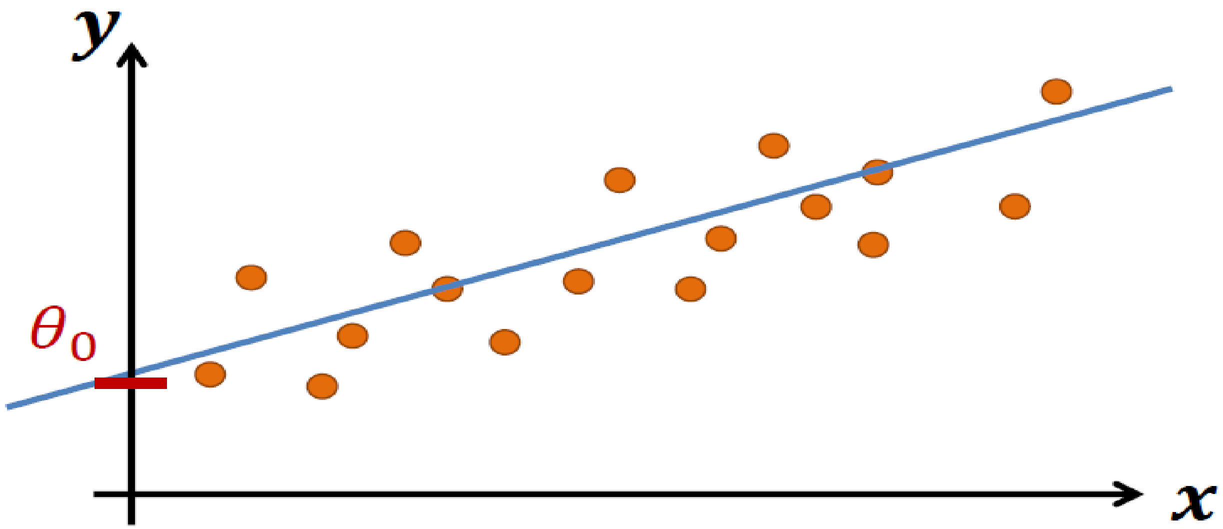
weights (i.e., model parameters)

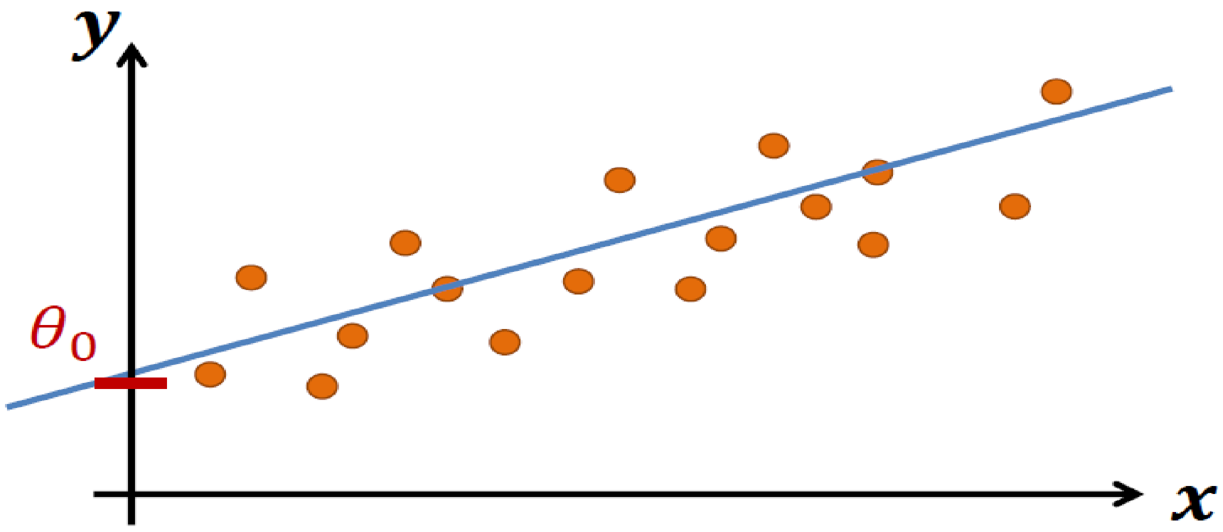
input data

Linear Prediction



Linear Prediction

$$\hat{y}_i = \theta_0 + \sum_{j=1}^{d} x_{ij}\theta_j$$



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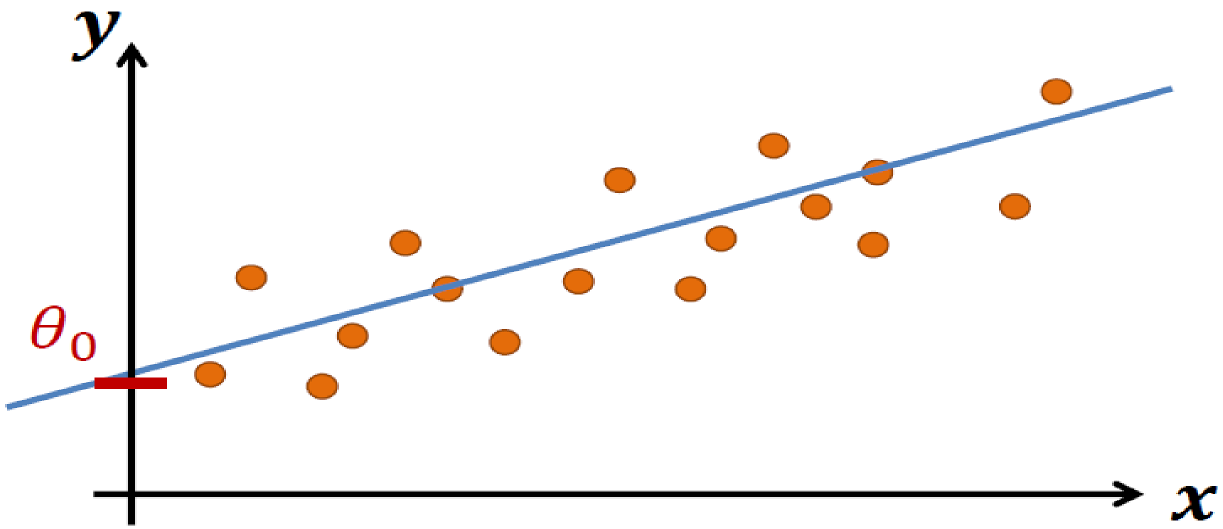
Linear Prediction

$$\hat{y}_i = \theta_0 + \sum_{j=1}^{d} x_{ij}\theta_j$$

bias or intercept term



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$$\hat{y}_i = \theta_0 + \sum_{j=1}^{d} x_{ij}\theta_j = \theta_0 + \begin{bmatrix} x_{i1} & \cdots & x_{id} \end{bmatrix} \begin{bmatrix} \theta_1 \\ \vdots \\ \theta_d \end{bmatrix}$$

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$$= \begin{bmatrix} 1 & x_{i1} & \cdots & x_{id} \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix}$$

Linear Prediction

Predictions

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix}$$

Linear Prediction

Design matrix
(one sample has d features)

Predictions

Model parameters
(d weights and 1 bias)

$$\begin{bmatrix} \hat{y}_1 \\ \hat{y}_2 \\ \vdots \\ \hat{y}_n \end{bmatrix} = \begin{bmatrix} 1 & x_{11} & \cdots & x_{1d} \\ 1 & x_{21} & \cdots & x_{2d} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{nd} \end{bmatrix} \begin{bmatrix} \theta_0 \\ \theta_1 \\ \vdots \\ \theta_d \end{bmatrix}$$

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$$\hat{\boldsymbol{y}} = \mathbf{X}\boldsymbol{\theta}$$

How to Obtain the Model?

Data points

$\mathbf{x}$

Labels (ground truth)

y

Model parameters

θ

Estimation

$\hat{y}$

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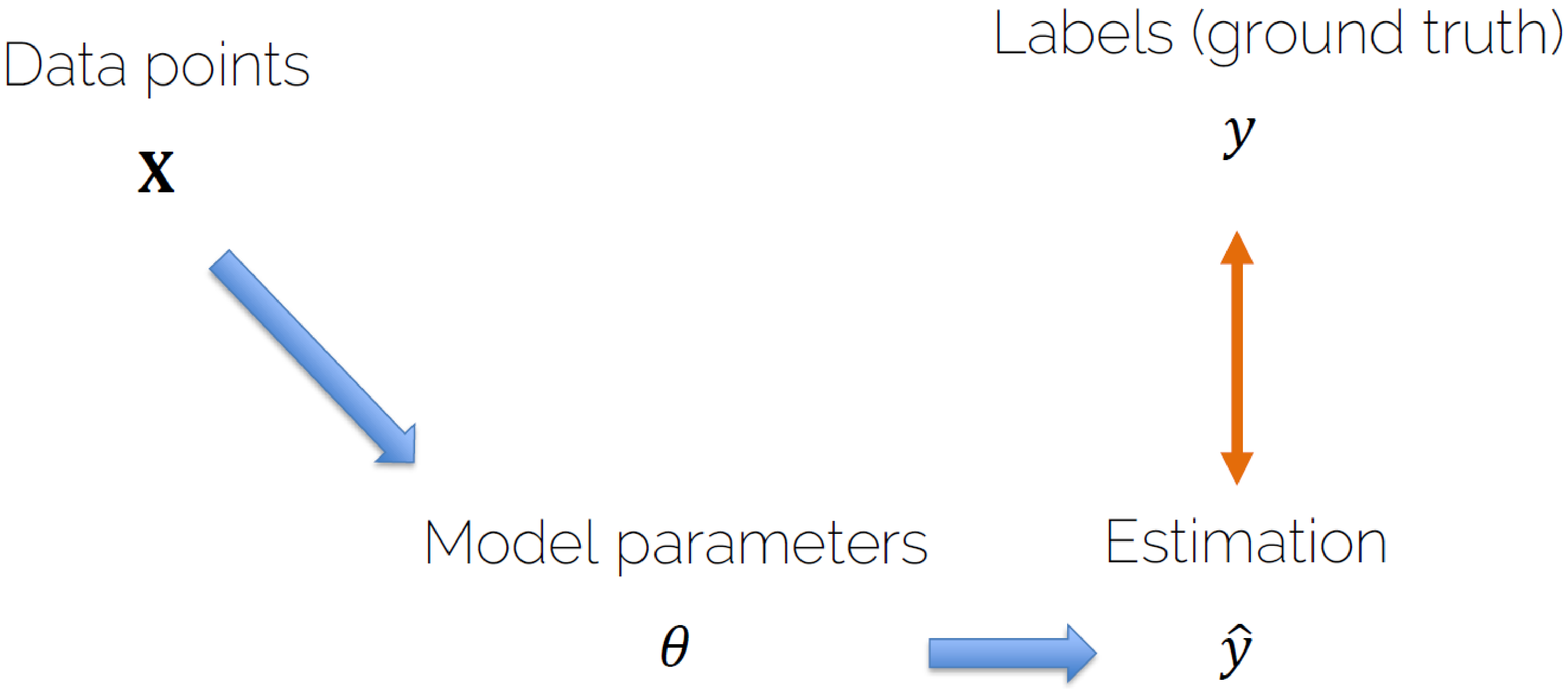

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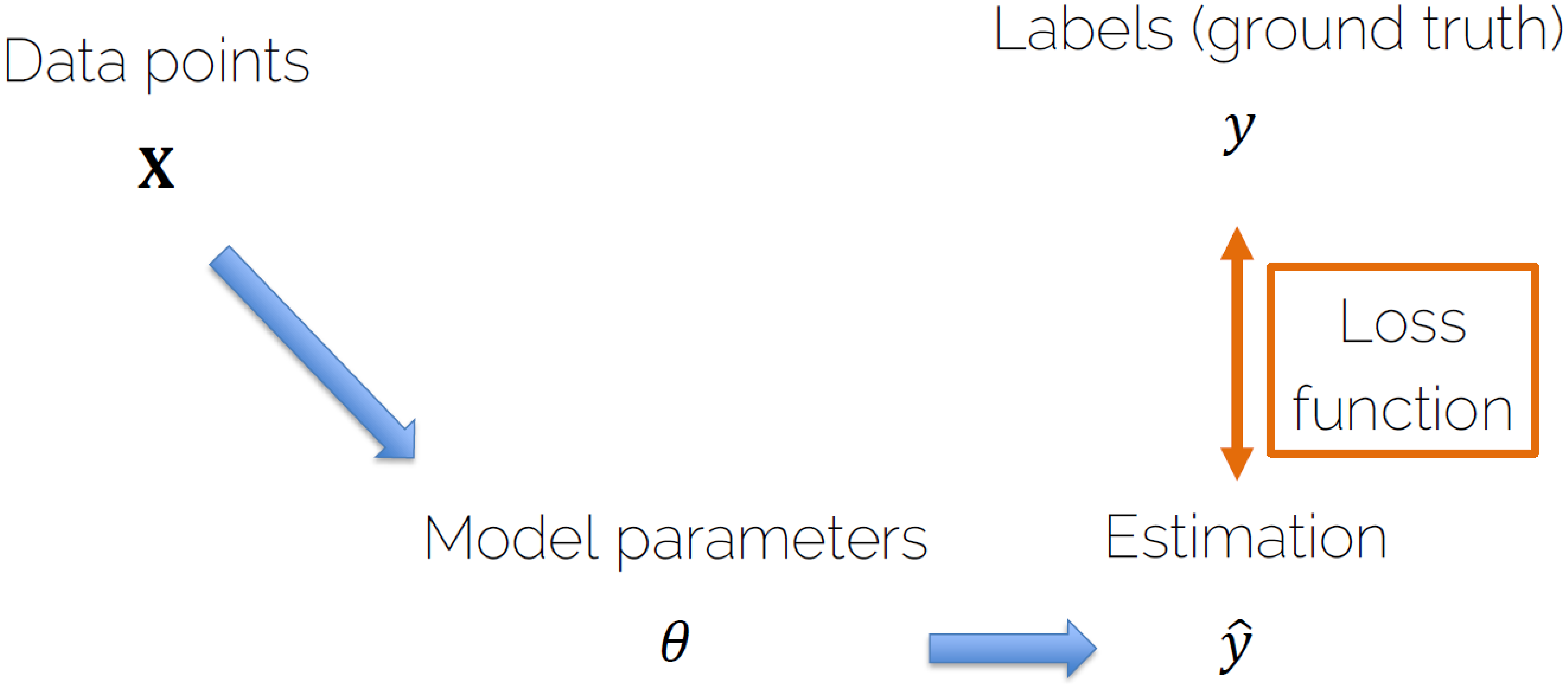

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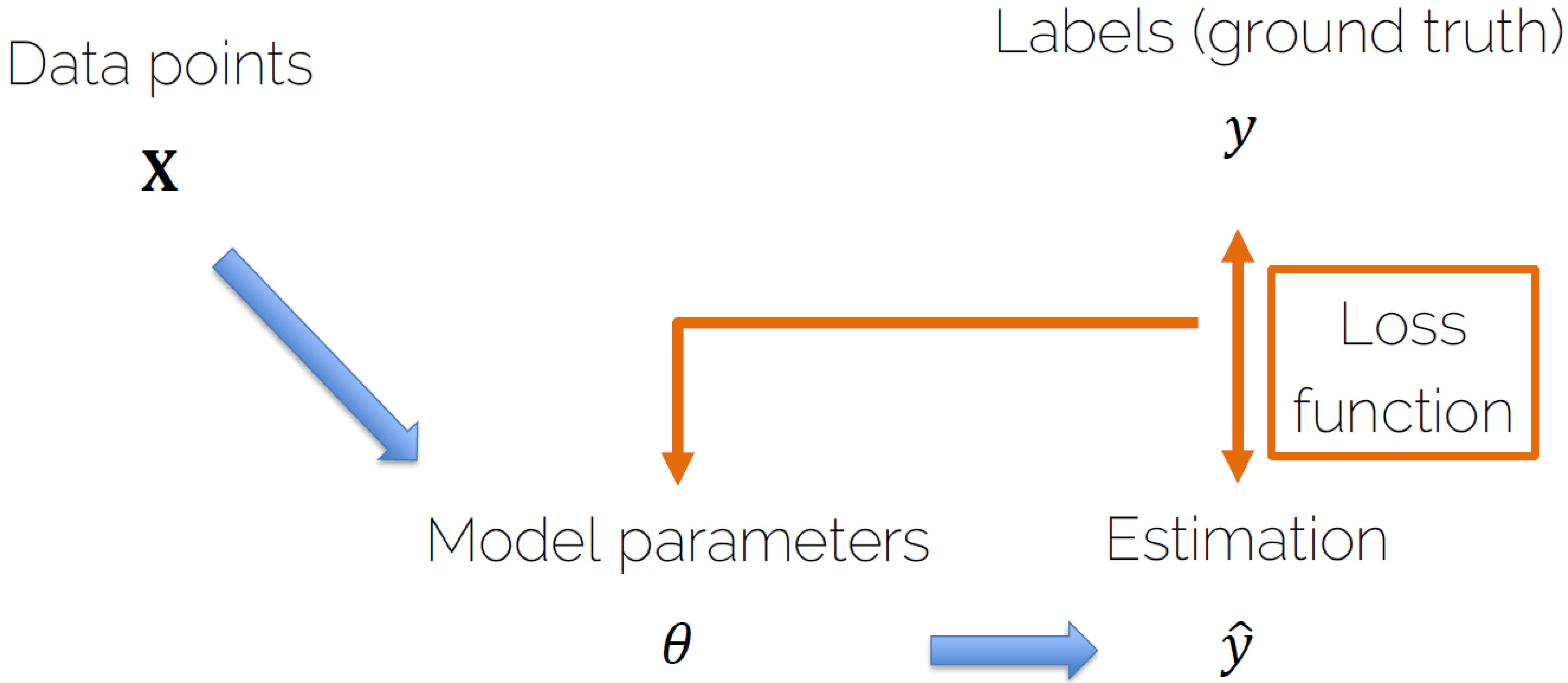

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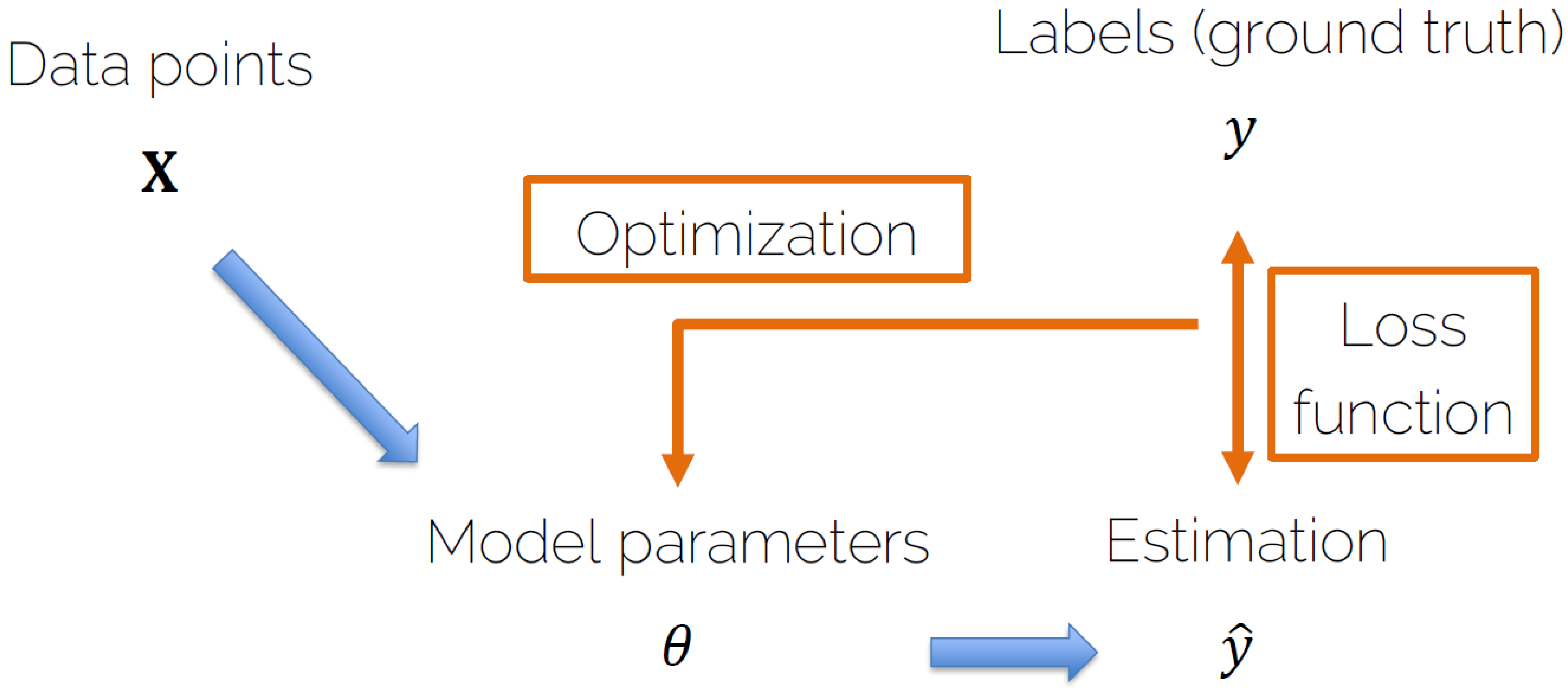

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How to Obtain the Model?

- **Loss function**: measures how good my estimation is (how good my model is) and tells the optimization method how to make it better.

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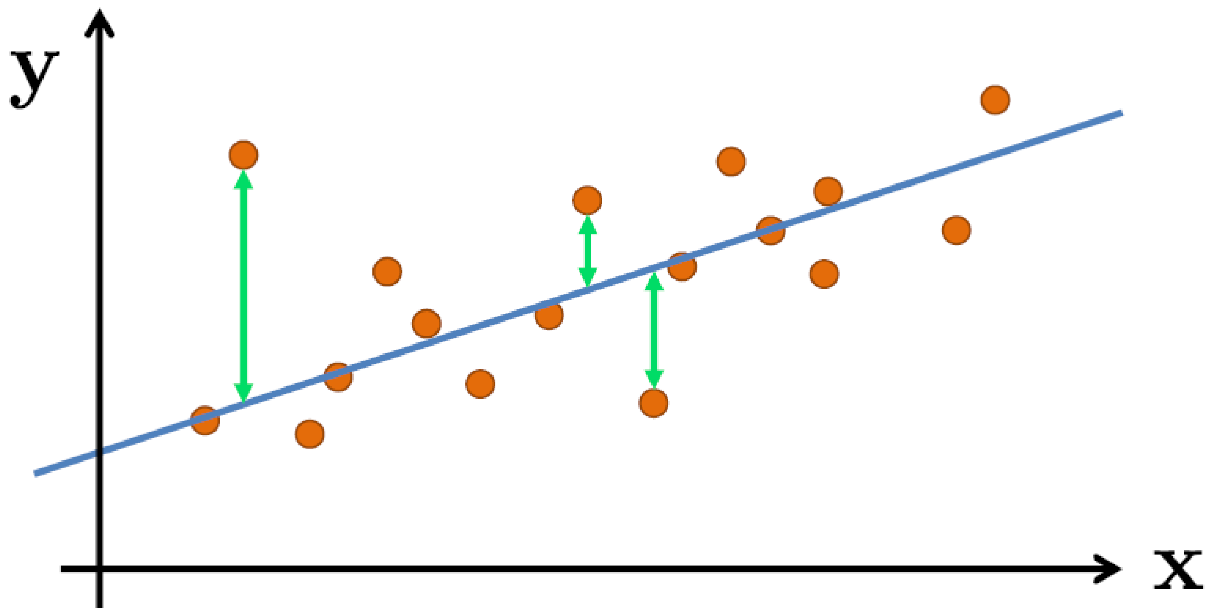
- **Optimization**: changes the model in order to improve the loss function (i.e., to improve my estimation).

Linear Regression



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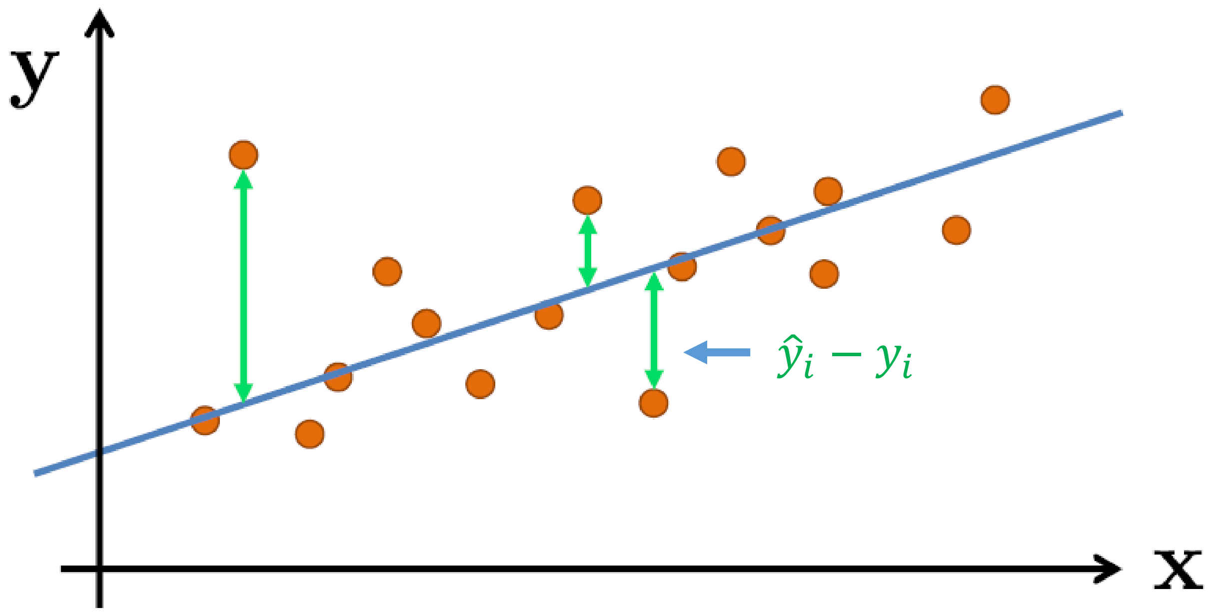
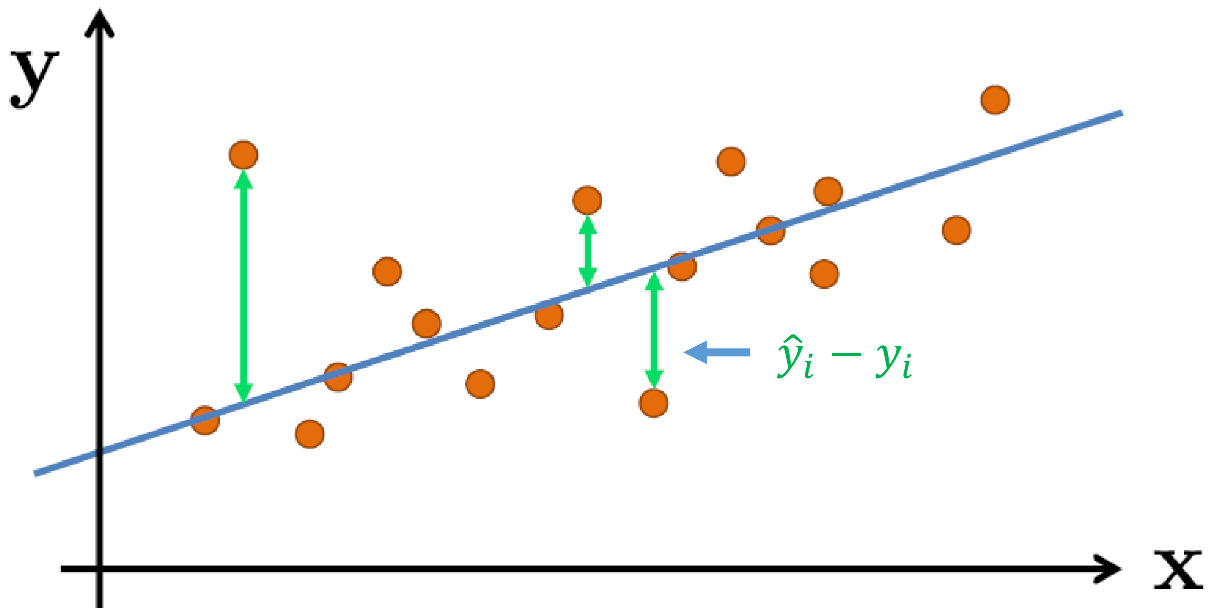


Cost function

$$J(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$

Linear Regression



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Optimization: Linear Least Squares

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Matrix notation

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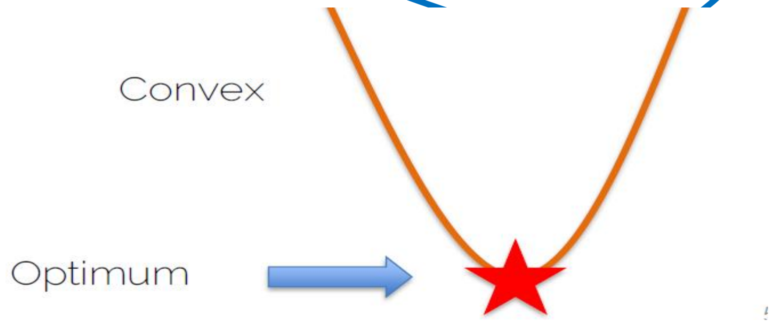
$$\frac{\partial J(\boldsymbol{\theta})}{\partial \boldsymbol{\theta}} = 0$$



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$$\boxed{\boldsymbol{\theta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{y}}$$

Analytical solution to a convex problem

Is this the best Estimate?

Least squares estimate

$$J(\boldsymbol{\theta}) = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$

Probabilistic Interpretation

Let us assume that the target variables and the inputs are related via the equation:

$$y^{(i)} = \theta^T x^{(i)} + \epsilon^{(i)}$$

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parameters (also called weights)

"input" variables (features)

error term that captures either unmodeled effects or random noise.

Probabilistic Interpretation

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Let us further assume that the error terms are independently and identically distributed (IID) according to a Gaussian distribution with mean zero and some variance σ^2

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Distribution of $y^{(i)}$ given $x^{(i)}$ and parametrized by θ

Probabilistic Interpretation

- Given $\mathbf{X}$ (the matrix containing all the $x^{(i)}$'s) and θ, what is the distribution of the $y^{(i)}$'s?

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- Given $\mathbf{X}$ (the matrix containing all the $x^{(i)}$'s) and θ, what is the distribution of the $y^{(i)}$'s?
- The probability of the data is given by $p(y|X;\theta)$. This quantity is typically viewed as a function of y (and perhaps $\mathbf{X}$) for a fixed value of θ.
- When we wish to explicitly view this as a function of θ, we will instead call it the **likelihood** function:

$$L(\theta) = L(\theta; X, y) = p(y|X;\theta)$$

Probabilistic Interpretation

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Note that by the independence assumption on the $\epsilon^{(i)}$ (and hence also the $y^{(i)}$'s given the $x^{(i)}$'s), this can also be written:

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$$L(\theta) = \prod_{i=1}^{m} p(y^{(i)} \mid x^{(i)}; \theta)$$

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$$\begin{aligned} L(\theta) &= \prod_{i=1}^{m} p(y^{(i)} \mid x^{(i)}; \theta) \\ &= \prod_{i=1}^{m} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right) \end{aligned}$$

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- We should choose θ to maximize $L(\theta)$.

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Instead of maximizing $L(\theta)$, we can also maximize any strictly increasing function of $L(\theta)$

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log likelihood $$\ell(\theta) = \log L(\theta)$$

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i.i.d. assumption

$$
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Logarithmic property
log(ab)=log$a$+log$b$

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Canceling log and e

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&= \log \prod_{i=1}^{m} \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right) \\
&= \sum_{i=1}^{m} \log \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(y^{(i)} - \theta^T x^{(i)})^2}{2\sigma^2}\right) \\
&= m \log \frac{1}{\sqrt{2\pi}\sigma} - \frac{1}{\sigma^2} \cdot \frac{1}{2} \sum_{i=1}^{m} (y^{(i)} - \theta^T x^{(i)})^2.
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Canceling log and e

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maximizing $\ell(\theta)$ gives the same answer as minimizing

Image Classification

Source: Niessner and Leal-Taixé, Introduction to Deep Learning

Regression vs Classification

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Regression vs Classification

- Regression:
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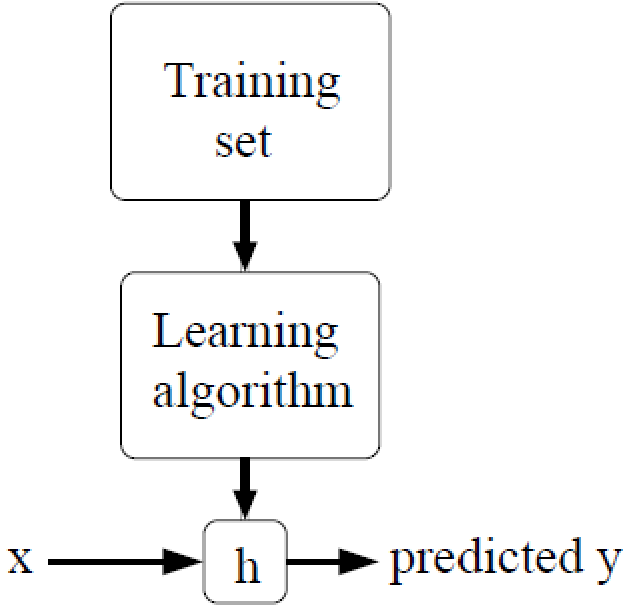
  - predict a discrete value
    - binary classification: output is either 0 or 1
    - multi class classification: set of N classes

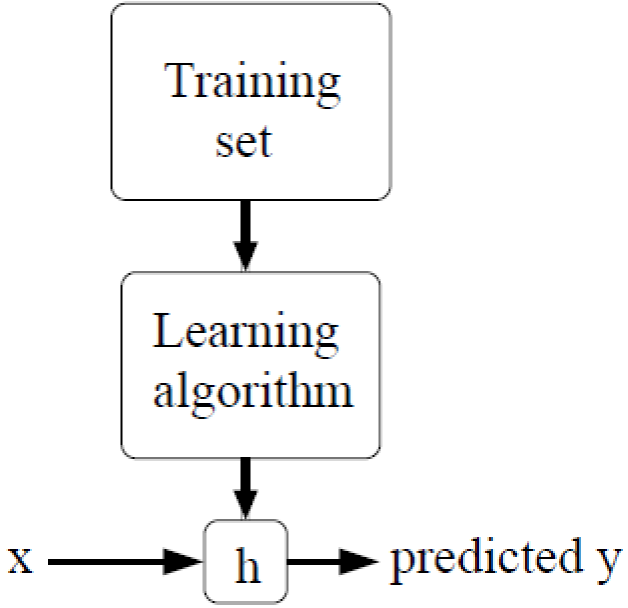
Logistic Regression

Supervised Learning



Adapted from: Andrew Ng, CS229: Machine Learning

Supervised Learning



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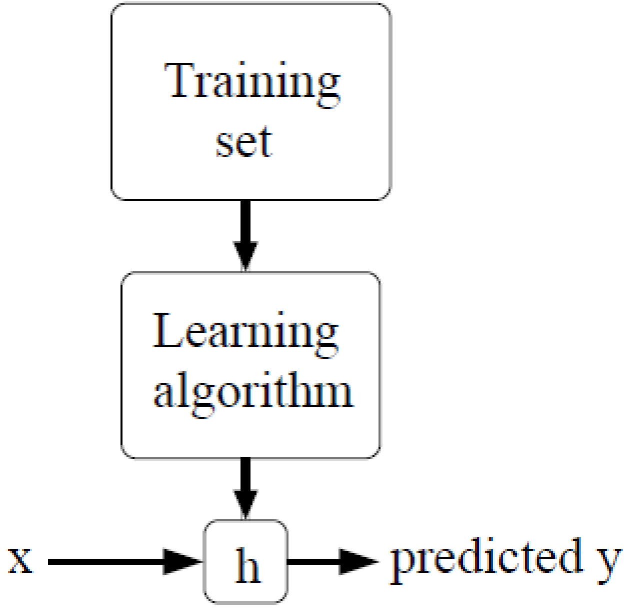

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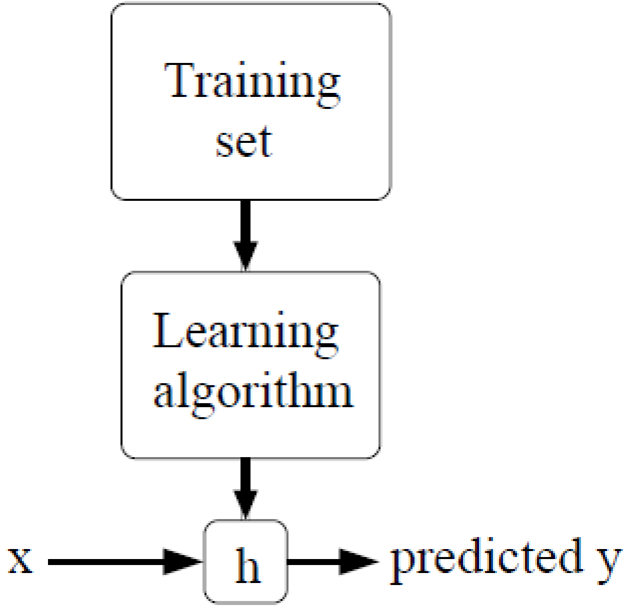

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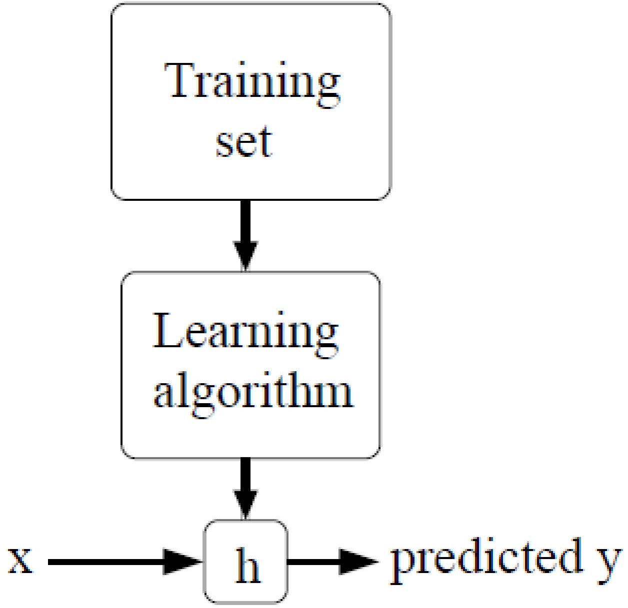

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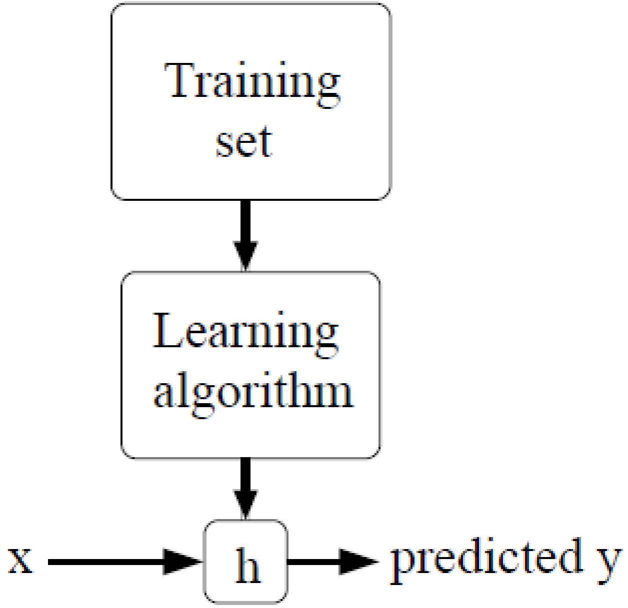

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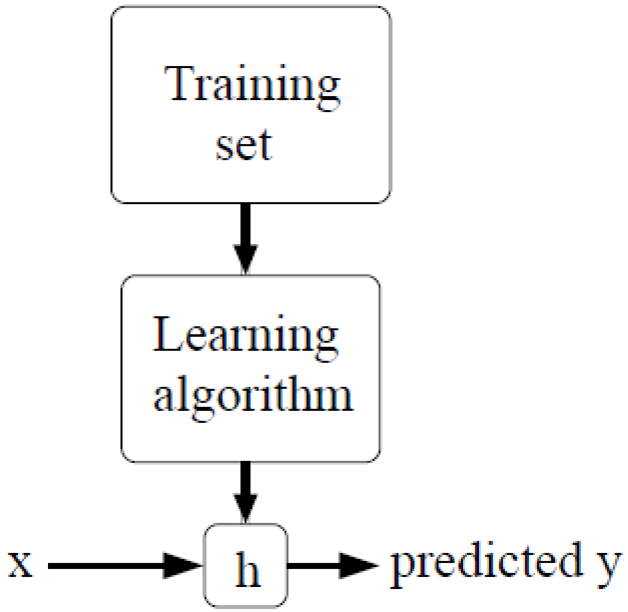

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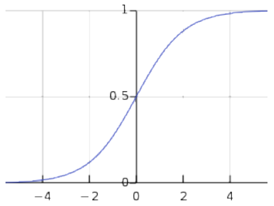
$$g(z) = \frac{1}{1 + e^{-z}}$$

sigmoid function



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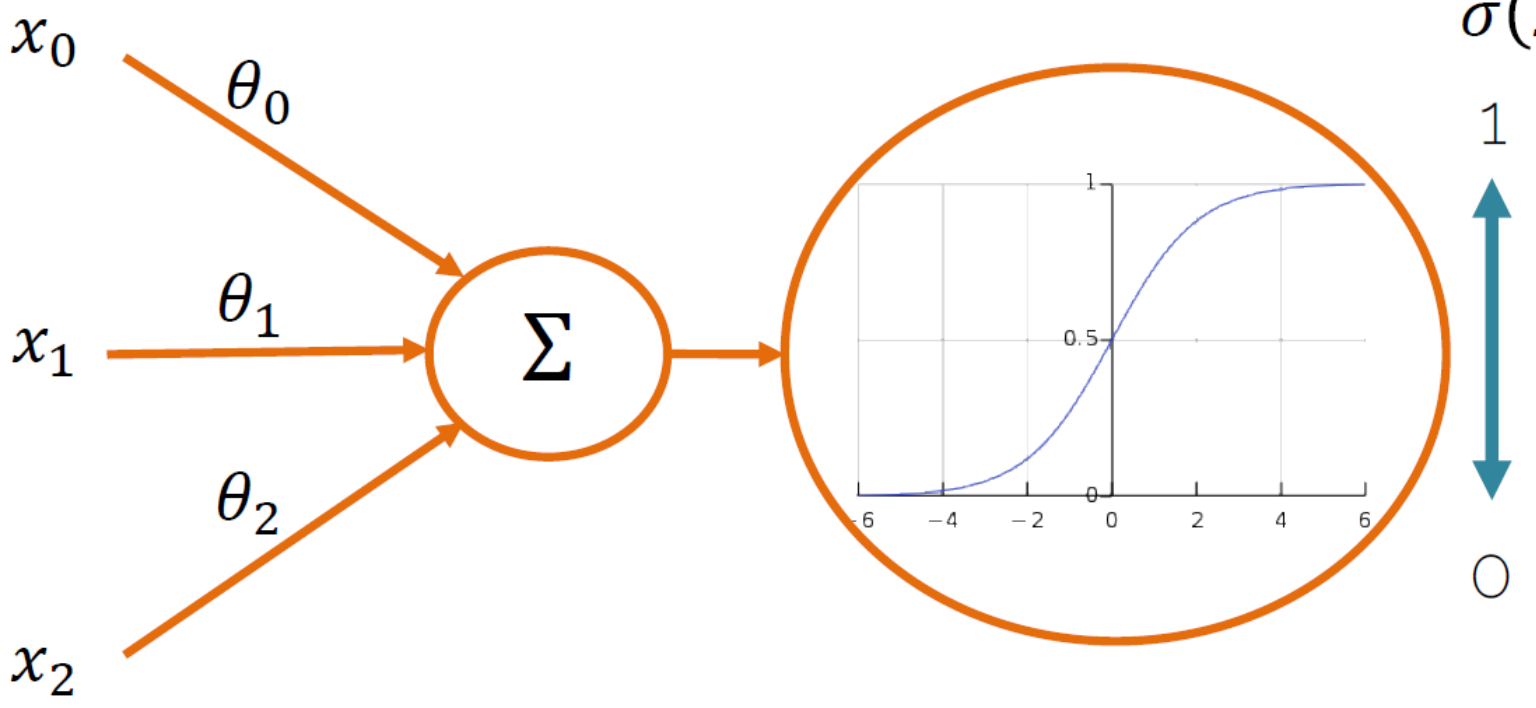
Sigmoid for Binary Predictions

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$



Can be interpreted as a probability

$$p(y_i = 1 | \mathbf{x}_i, \boldsymbol{\theta})$$

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This can be written more compactly as:

$$p(y \mid x; \theta) = (h_\theta(x))^y \left(1 - h_\theta(x)\right)^{1-y}$$

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$$\begin{aligned} \ell(\theta) &= \log L(\theta) \\ &= \sum_{i=1}^{m} y^{(i)} \log h(x^{(i)}) + (1 - y^{(i)}) \log(1 - h(x^{(i)})) \end{aligned}$$

Logistic Regression: Optimization

Loss function

$$L(\hat{y}_i, y_i) = -(y_i \log \hat{y}_i + (1 - y_i)\log(1 - \hat{y}_i))$$

Cost function

$$C(\theta) = \frac{1}{n}\sum_{i=1}^{n} L(\hat{y}_i, y_i)$$